

EKS Architectural Mastery

Technical Reference: Kubernetes on AWS

1. The Managed Controller Model

In EKS, AWS manages the control plane (API Server, etcd, Scheduler) as a service. This eliminates the burden of updating master nodes, allowing engineers to focus on the Data Plane (Worker Nodes or Fargate).

2. AWS VPC CNI (Native Networking)

Unlike overlay-reliant networks, the AWS VPC CNI makes Pods first-class citizens in the VPC. This enables native flow logs and security group integration.

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# How VPC CNI Works:  
# 1. EC2 instances host multiple Elastic Network Interfaces (ENIs).  
# 2. VPC CNI allocates secondary IP addresses from ENIs.  
# 3. Each Pod receives one of these native VPC IPs.
```

3. Authentication: IRSA & OIDC

The standard for security is IRSA (IAM Roles for Service Accounts). By annotating a Service Account, Pods obtain temporary, scoped AWS credentials via OIDC.

Security Architecture: Never grant administrative IAM roles to the Cluster Worker Nodes. Use IRSA to adhere to the Principle of Least Privilege, giving only specific Pods access to resources like S3 or DynamoDB.

4. AWS Load Balancer Controller

This controller watches for Kubernetes `Ingress` or `Service` resources, automatically provisioning ALB (Application Load Balancers) for path-based routing or NLB (Network Load Balancers) for static IP/High throughput needs.

5. Storage: EBS CSI Driver

EBS CSI enables snapshots, resizing, and encryption. Note: `volumeBindingMode: WaitForFirstConsumer` is essential to ensure a volume is provisioned in the same Availability Zone where the pod is scheduled.